

A Benchmarking Case Study for Local Flexibility Markets: Network, Scenarios, and Open Inputs

Savvas Panagi , *Student Member, IEEE*, Chrysovalantis Spanias , *Member, IEEE*, and Petros Aristidou , *Senior Member, IEEE*

Abstract—External inputs to local flexibility markets, network models, demand scenarios, flexibility offers, and asset thermal, economic parameters do not change the clearing methodology, yet they are routinely defined under heterogeneous assumptions, which hinders reproducible benchmarking. This paper provides a complete, methodology-agnostic input set built around a modified CIGRE MV network with prescribed base-load and stress-load conditions, synthetic flexibility offers, wholesale prices, and transformer and cable parameters. Baseline AC operating points (voltages, loadings, and aging) are reported for both days so that alternative flexibility-request and clearing methods can be applied to the same inputs and compared fairly. The associated data are released openly on Zenodo.

Index Terms—Local flexibility markets, benchmarking, CIGRE MV network, open data, transformer aging, cable aging.

I. INTRODUCTION

Local flexibility markets (LFMs) separate, to varying degrees, the market-clearing mechanism from the network-specific inputs that drive flexibility valuation. The clearing formulation itself, objective structure, offer acceptance rules, and privacy constraints, can be studied independently of any particular feeder. In contrast, the *external inputs* that feed the clearing problem, including the network model, baseline and stressed demand profiles, flexibility offers, wholesale prices, and the thermal and economic parameters of transformers and cables, do not alter the methodology but strongly affect numerical outcomes. Across the literature these inputs are typically assembled under heterogeneous and often incompletely documented assumptions, which limits reproducibility and makes fair benchmarking between flexibility-request methods difficult.

This paper addresses that gap by defining a complete, openly specified input set for LFM benchmarking. We adopt a modified CIGRE European MV network, prescribe base-load and stress-load (EV) operating conditions, publish synthetic but fully stated flexibility offers, and fix ambient temperature together with transformer and cable thermal-economic parameters. All network data, time series, and parameter files are released in an open-access Zenodo repository [1]. Baseline AC power-flow results (voltages, loadings, and aging) are reported without assuming any particular clearing method,

S. Panagi and C. Spanias are with the Distribution System Operator, Electricity Authority of Cyprus, Nicosia, Cyprus.

S. Panagi and P. Aristidou are with the Department of Electrical & Computer Engineering & Informatics, Cyprus University of Technology, Limassol, Cyprus.

Network data, load profiles, flexibility offers, and parameter files are available in an open-access Zenodo repository [1].

so that independent groups can apply their own flexibility-request or market-clearing formulations to the same inputs and compare outcomes on equal footing.

The remainder of the paper is organized as follows. Section II specifies the network, scenarios, and numerical inputs. Section III reports the baseline operating points for the base and stress days. Section IV describes the Zenodo release. Section V concludes.

II. CASE STUDY: NETWORK, SCENARIOS, AND INPUTS

This section defines the network, operating scenarios, and numerical inputs used for benchmarking. The same specification underlies the numerical experiments reported in [2]; the machine-readable files are released on Zenodo [1].

A. Network Model and Modifications

The case study is built on the standard CIGRE European MV benchmark network as depicted in Fig. 1. The original model represents a two-feeder meshed 110/20 kV system with aggregated MV loads, although the network is operated

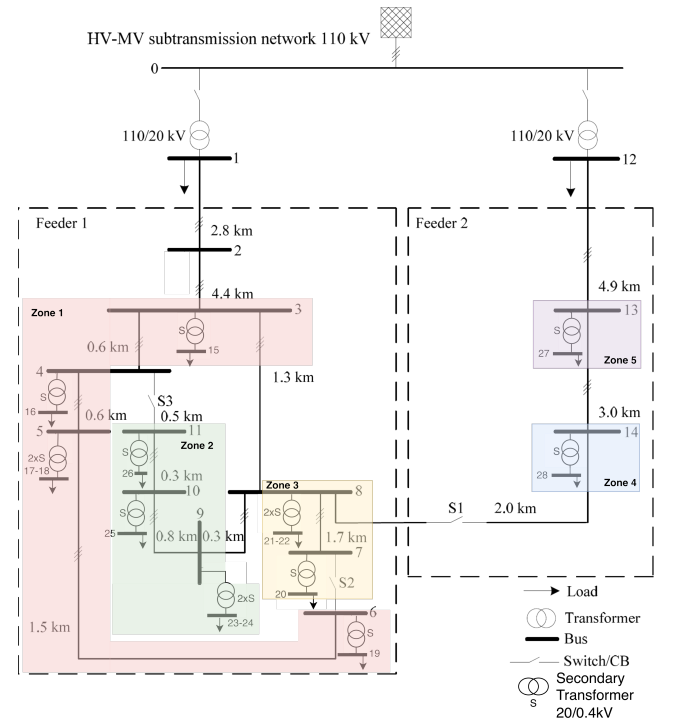


Fig. 1. CIGRE single-line diagram of the studied 20-kV distribution system, including the primary and secondary substations.

radially, with the tie switches open. For the present study, the network is modified such that each controllable MV load point is represented by a dedicated 20/0.4 kV secondary substation. Two MV buses, buses 1 and 12, are excluded from this secondary-substation expansion. These buses represent large, non-controllable MV demand points directly connected to the two primary feeders and remain fixed throughout the study.

Following these modifications, the network contains two primary transformers and 14 secondary substations, as shown in Fig. 1. All simulations are performed over a 24-hour horizon with hourly time steps. Because some flexibility-request methodologies partition buses into zones in order to enlarge the set of eligible providers and thereby increase market liquidity, e.g., [3], the benchmarking setup also defines five indicative zones as illustrated in Fig. 1.

B. Operating Scenarios

Two operating scenarios are considered. The *base case* represents a typical suburban load day. A common normalized *duck-curve* active-load profile (in p.u.) is applied identically to every load connected to the network; the resulting primary- and secondary-transformer feeder active-power trajectories are shown in Fig. 2. Because all loads share the same shape, the feeder time series differ only through their rated demand levels. The *stress case* retains this duck-curve base demand but superimposes uncontrolled residential EV charging at each secondary substation. The EV load follows the statistical assumptions used in [4]. Each secondary substation serves 30–50 customers based on their nominal apparent power, of which 30% own an EV. Connection type is assigned probabilistically, with 80% to three-phase supply and 20% to single-phase supply. The corresponding home chargers are rated at 11 kW (three-phase) and 7 kW (single-phase). Each EV charges at its rated charger power until the daily trip energy is restored. Fig. 3 shows the resulting EV active-power consumption at each secondary and the relative increase in secondary-transformer loading, which is concentrated in the evening charging window.

C. Input Parameters

1) *Transformer aging parameters*: Transformer aging follows the IEEE C57.91 Arrhenius model [5]. The rated temperature rises are $\Delta\theta_R^{TO} = 55^\circ\text{C}$ and $\Delta\theta_R^H = 25^\circ\text{C}$ [6]. Oil

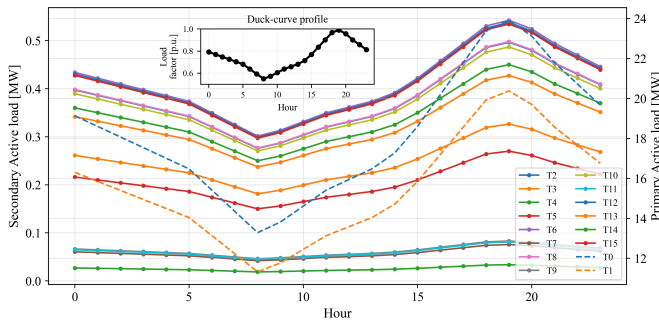


Fig. 2. Base-case demand: common duck-curve load factor and the corresponding active-power profiles at the primary and secondary transformer feeders.

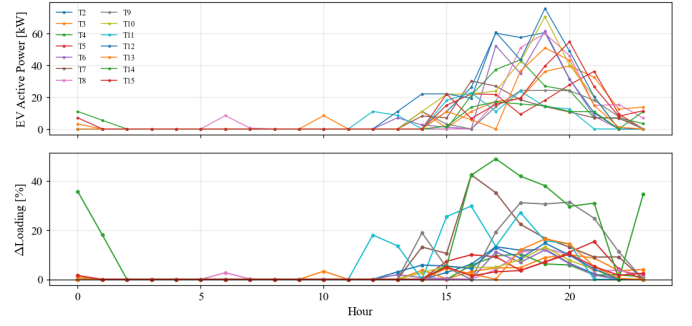


Fig. 3. Stress case: uncontrolled EV active power at each secondary substation (top) and the resulting relative secondary-transformer loading increase $\Delta\text{Loading}$ relative to the base case (bottom).

and winding exponents are $n = m = 0.8$ [5]; the top-oil and hotspot time constants are 3 h and 4 min [6], respectively. The normal life expectancy is taken as 180 000 h [5], and the replacement cost is assumed proportional to rated power at 100 000 EUR/MVA [7].

2) *Cable aging parameters*: Cable aging uses an Arrhenius conductor-temperature model with parameters from [6, 8]: rated surface and core temperature rises $\Delta\theta_{CS,R} = 38^\circ\text{C}$ and $\Delta\theta_{C,R} = 40^\circ\text{C}$, time constants $\tau_{CS} = 2$ h and $\tau_C = 4$ min, Arrhenius coefficient $B = 15,000$, normal cable life 438 000 h, and replacement cost 80 000 EUR/km multiplied by the installed cable length.

3) *Market and network assumptions*: The remaining inputs are network limits and exogenous price/offer data. Voltage limits are $\underline{V} = 0.90$ p.u. and $\bar{V} = 1.10$ p.u. The baseline AC operating point assumes a constant load power factor of 0.95 (lagging). Wholesale energy prices follow the Cyprus day-ahead clearing price on 18 July 2026 [9]. Fourteen local FSPs (one per secondary substation) submit synthetic hourly offers with capacities of 120 kW and bid prices drawn uniformly from 60–90 €/MWh; the wholesale series and the resulting bid-price heatmap are shown in Fig. 4. The inputs are summarized in Table I.

III. BASELINE OPERATING-POINT RESULTS

Given the network, demand scenarios, and asset parameters of Section II, a full AC power flow is solved independently for each hour of the base and stress days. No flexibility activation is considered in this section: the purpose is to document the

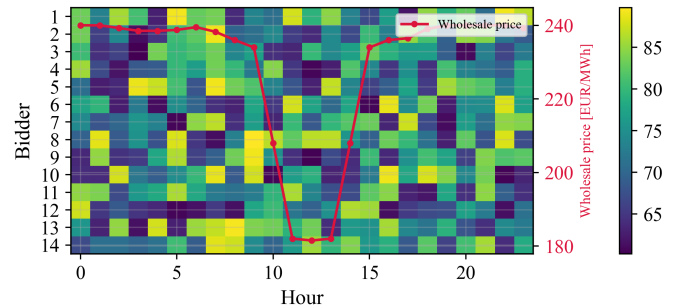


Fig. 4. Wholesale day-ahead price and FSP hourly bid prices by bidder.

TABLE I
SUMMARY OF KEY BENCHMARKING PARAMETERS.

Quantity	Value
Horizon / resolution	24 h / 1 h
Primary transformers	2
Secondary substations / FSPs	14 / 14
FSP capacity (each)	120 kW
FSP bid-price range	60–90 €/MWh
Load power factor	0.95 (lagging)
Voltage limits	0.90–1.10 p.u.
Transformer replacement cost	100 000 EUR/MVA
Cable replacement cost	80 000 EUR/km

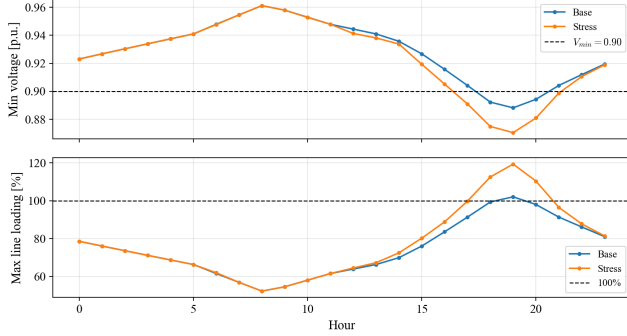


Fig. 5. Baseline AC operating point: minimum feeder voltage (top) and maximum line loading (bottom) for the base and stress cases. Dashed lines mark $\underline{V} = 0.90$ p.u. and 100% loading.

resulting operating points so that any flexibility-request or clearing methodology can be applied to the same inputs and compared on equal footing.

A. Voltage and thermal loading

Fig. 5 compares the network-wide minimum bus voltage and the maximum line loading over the day. In the base case, the evening peak drives the minimum voltage slightly below $\underline{V} = 0.90$ p.u. at hours 18–20 (lowest value ≈ 0.89 p.u. at hour 19), while the maximum line loading exceeds 100% only at hour 19 ($\approx 102\%$). Under the stress case, uncontrolled EV charging deepens and prolongs these violations: the minimum voltage falls to ≈ 0.87 p.u. at hour 19 and remains below 0.90 p.u. over hours 17–21, and the maximum line loading reaches $\approx 119\%$ at hour 19, with overload over hours 18–20.

B. Transformer and cable aging at the baseline

With the thermal-economic parameters of Section II, the exact nonlinear aging models are evaluated on the baseline loadings $K_{tra,t}$ and $K_{l,t}$ (no flexibility activation). Fig. 6 shows primary transformer T0 (25 MVA): the peak hot-spot temperature rises from 123.1°C (base) to 125.2°C (stress), the peak FAA from 3.64 to 4.47, and the daily aging cost from 446 to 488 € (+9.4%). Fig. 7 reports the most loaded MV cable under stress (line 9, 1.3 km): conductor temperature reaches 120.6°C versus 95.6°C in the base day, peak FAA jumps from 1.88 to 24.8, and the daily aging cost from 1.5 to 11.5 €. Aggregating over all MV lines, the daily cable aging cost increases from about 10 to 49 €. These

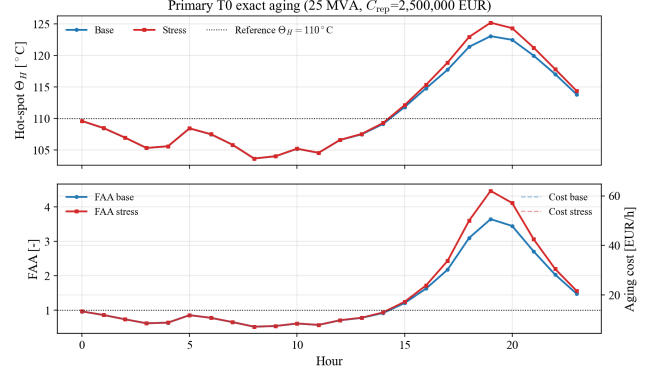


Fig. 6. Exact IEEE C57.91 aging of primary transformer T0 under the base and stress baselines: hot-spot temperature (top) and FAA with hourly aging cost (bottom).

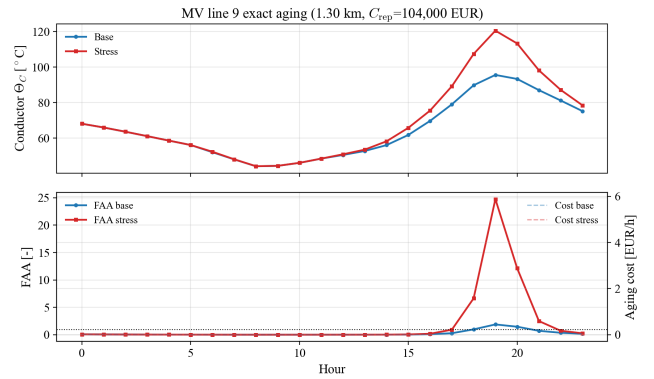


Fig. 7. Exact transient Arrhenius aging of the most loaded MV cable (line 9) under the base and stress baselines: conductor temperature (top) and FAA with hourly aging cost (bottom).

trajectories provide a shared numerical reference for methods that monetize asset degradation on this benchmark.

IV. DATA AVAILABILITY

The complete benchmarking package, including the modified CIGRE network in a pandapower format, base- and stress-load profiles, flexibility-offer sets, wholesale price series, and transformer/cable thermal and economic parameters, is available under open access on Zenodo [1]. The repository is intended to support reproducible comparison of LFM methods under a shared external-input specification.

V. CONCLUSION

This paper provides an openly specified benchmarking package for local flexibility markets: a modified CIGRE MV network, base and stress demand scenarios, FSP offers, wholesale prices, and transformer/cable thermal-economic parameters, together with the associated baseline AC operating points (voltages, loadings, and aging). Because the package is methodology-agnostic, alternative flexibility-request and clearing formulations can be applied to the same inputs and compared fairly. All data are released on Zenodo [1] to facilitate reuse and extension.

REFERENCES

- [1] S. Panagi, C. Spanias, and P. Aristidou, “Benchmarking case study for local flexibility markets: Network, scenarios, and open inputs,” Zenodo, 2026. [Online]. Available: <https://doi.org/10.5281/zenodo.21775990>
- [2] —, “Dynamic flexibility requests in local flexibility markets: Quantifying the DSO willingness to pay,” *arXiv preprint*, 2026.
- [3] E. Prat, I. Dukovska, R. Nellikkath, M. Thoma, L. Herre, and S. Chatzivasileiadis, “Network-aware flexibility requests for distribution-level flexibility markets,” *IEEE Transactions on Power Systems*, vol. 39, no. 2, pp. 2641–2652, 2024.
- [4] S. Panagi, P. Therapontos, C. Spanias, and P. Aristidou, “Impact of uncontrolled electric vehicle charging on unbalanced suburban low-voltage networks,” in *2025 IEEE PES Innovative Smart Grid Technologies Conference Europe (ISGT Europe)*, 2025, pp. 1–5.
- [5] “Ieee guide for loading mineral-oil-immersed transformers and step-voltage regulators,” *IEEE Std C57.91-2011*, pp. 1–123, 2012.
- [6] P. Andrianesis, A. Michiorri, G. Kariniotakis, and M. Caramanis, “Impact of transformer and cable aging on distribution locational marginal costs in active distribution networks,” in *IET Conference Proceedings CP785*, no. 6, 2021, pp. 2237–2241.
- [7] Balducci, Schienbein, Nguyen, Brown, and Fathelrahman, “An examination of the costs and critical characteristics of electric utility distribution system capacity enhancement projects,” in *2005/2006 IEEE/PES Transmission and Distribution Conference and Exhibition*, 2006, pp. 78–86.
- [8] E. N. Association *et al.*, “Climate change adaptation reporting power second round,” 2015.
- [9] Transmission System Operator of Cyprus, “Day-ahead market clearing prices,” <https://tsoc.org.cy/competitive-electricity-market/dam-volume-prices-graph/>, 2026, accessed: Jul. 17, 2026.